

Day: Tuesday

16018

Date: 12/5/26

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Submitted To : Mr. Mohsin

Subject : PS

ID : 16018

Date : 12-5-26

— ~~CE~~ ANSWER 1 ~~BD~~ —

- Number of trials ( $n$ ): 1
- Number of successes ( $k$ ): 1
- Probability of success ( $p$ ):  $3/8$
- Probability of failure ( $q$ ):  $1 - 3/8 = 5/8$

Apply Formula:-

$$P(X=1) = \binom{1}{1} \left(\frac{3}{8}\right)^1 \left(\frac{5}{8}\right)^{1-1}$$

$$P(X=1) = 1 \times \frac{3}{8} \times 1 = \frac{3}{8}$$

$$P(X=1) = \frac{3}{8}$$

— ~~CE~~ ANSWER 2 ~~BD~~ —

- Number of trials ( $n$ ): 1
- Number of successes ( $k$ ): 1
- Probability of success ( $p$ ):  $6/36 = 1/6$
- Probability of failure ( $q$ ):  $1 - 1/6 = 5/6$

Apply Formula:-

$$P(X=1) = \binom{1}{1} \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^{1-1}$$

$$P(X=1) = 1 \times \frac{1}{6} \times 1$$

$$P(X=1) = \frac{1}{6}$$



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— ANSWER 3 —

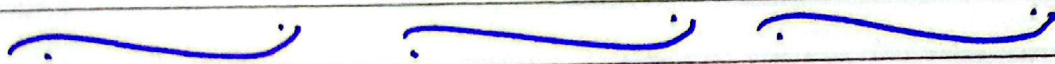
- Number of trials ( $n$ ): 1
- Number of successes ( $k$ ): 1
- Probability of success ( $p$ ):  $2/4 = 1/2$
- Probability of failure ( $q$ ):  $1 - 1/2 = 1/2$

Apply Formula:-

$$P(X=1) = \binom{1}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{1-1}$$

$$P(X=1) = 1 \times \frac{1}{2} \times 1$$

$$P(X=1) = \frac{1}{2}$$



— ANSWER 4 —

- Number of trials ( $n$ ): 1
- Number of successes ( $k$ ): 1
- Probability of success ( $p$ ):  $1/4$
- Probability of failure ( $q$ ):  $1 - 1/4 = 3/4$

Apply Formula:-

$$P(X=1) = \binom{1}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^{1-1}$$

$$P(X=1) = 1 \times \frac{1}{4} \times 1$$

$$P(X=1) = \frac{1}{4}$$

